

Bachelor of Technology — General Engineering - Open Specialisation

Table 1: General Engineering - Open Specialisation — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	40
	Discipline Electives	26
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech General Engineering - Open Specialisation — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical Systems Around Us	3-0-0-3	3
2	IC-241	Material Science for Engineers	3-0-0-3	3
3	ME-212	Product Manufacturing Technology	2-0-3-3	3
4	ME-213	Thermodynamics	3-1-0-4	3
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	HS-307	Macroeconomics	3-0-0-3	3
8	IC-253	Programming and Data Structures	2-0-2-3	3
9	ME-100	Reverse Engineering	0-0-2-1	1
10	DS-201	Data Handling and Visualisation	2-0-2-3	3
11	EE-231	Measurement and Instrumentation	2-0-2-3	3
12	EE-302	Control Systems	3-0-2-4	4
13	ME-305	Design of Machine Elements	3-1-0-4	4
14	HS-541	Technical Communication	1-0-0-1	1
Total				40

Table 3: B. Tech General Engineering - Open Specialisation — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech General Engineering - Open Specialisation — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech General Engineering - Open Specialisation — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
4	DC	IC-241	Material Science for Engineers	3-0-0-3	3
5	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
6	DC	ME-213	Thermodynamics	3-1-0-4	3
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech General Engineering - Open Specialisation — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-211	Analog Circuit Design	2-0-2-3	3
4	DC	HS-307	Macroeconomics	3-0-0-3	3
5	DC	IC-253	Programming and Data Structures	2-0-2-3	3
6	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech General Engineering - Open Specialisation — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-201	Data Handling and Visualisation	2-0-2-3	3
2	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
3	DC	EE-302	Control Systems	3-0-2-4	4
4	DC	ME-305	Design of Machine Elements	3-1-0-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech General Engineering - Open Specialisation — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-541	Technical Communication	1-0-0-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech General Engineering - Open Specialisation — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech General Engineering - Open Specialisation — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 26 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: General Engineering - Open Specialisation — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-502	Advanced Design Practicum	4
2	AR-505	Principles of Robot Autonomy	3
3	AR-519	Robot Manipulators	3
4	AR-521	Control of Robotic Systems	3
5	AR-522	Robot Vision	3
6	AR-526	Multimodal Learning	3
7	CS-212	Design and Analysis of Algorithms	4
8	CS-313	Computer Networks	4
9	CS-315	Artificial Intelligence (CS-362, CS-562)	4
10	CS-514	Advance Data Structure and Algorithm (CS-580)	4

General Engineering - Open Specialisation — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
11	DS-314	Probability Theory and Statistics	4
12	EE-203	Network Theory	3
13	EE-260	Signals and Systems	3
14	EE-301	Control Systems	3
15	EE-311	Device Electronics for Integrated Circuits	3
16	EE-503	Advance Communication Theory	3
17	EE-517	Wireless Communications and Networking	3
18	EE-529	Embedded Systems	4
19	EE-533	Acess Network	4
20	EE-536	IoT Systems	3
21	EE-607	Optical Communication Systems	3
22	EE-608	Digital Image Processing	4
23	EE-641	Advance Wireless Technologies	4
24	MA-546	Introduction to Quantitative Finance	4
25	MA-551	Numerical Analysis	3
26	ME-210	Fluid Mechanics	3
27	ME-452	Robot Kinematics, Dynamics and Control	4
28	ME-510	Advanced Manufacturing Processes	3
<i>Compulsory project components — counted as DE, not selectable</i>			
29	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
30	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4